

Hojun You

hjyou9332@gmail.com ◇ 346-459-5904 ◇ linkedin.com/in/hojun-you

PROFESSIONAL EXPERIENCE

Line Plus Corporation

Seongnam, South Korea

Machine Learning Engineer

09/2024 - Present

- Developed and deployed end-to-end computer vision models (AuraFace) for celebrity recognition in short-form videos, enhancing recommendation systems. (python, docker, airflow, git)
- Optimized video search retrieval for multilingual text queries by combining Qwen2.5-VL and llama-based embeddings with VideoCLIP(v3) embeddings, improving top 5 search accuracy by 40%. Integrated celebrity recognition models, enabling celebrity search in video content without explicit tagging.
- Devised a video moderation model based on VLM (InternVL3), improving moderation accuracy by X% and reducing costs by 80% (on progress).

RESEARCH EXPERIENCE

University of Houston

Houston, TX

Postdoctoral Researcher

09/2022 - 08/2024

- Analyzed user click behaviors on e-commerce websites (Amazon and Taobao) with encoder-only Transformer. Enhanced explanations of inter-category product relationships and search prediction accuracy by 20%. Applied the same model to predict user retweet patterns (likelihood increased by 60%). (python, hpc)
- Enhanced spatiotemporal terrorism prediction using an extended RNN & Transformer framework, reducing computational time by 10x (working paper).
- Boosted protein property prediction accuracy by 20% via BERT-based transfer learning.
- Improved heavy-tailed rainfall prediction accuracy by 60% using neural networks. Leveraged explainable AI techniques (SHAP and permutation importance) to identify key features, aligning results with scientific insights (link).

Seoul National University

Seoul, South Korea

Postdoctoral Researcher

03/2022 - 08/2022

Graduate Researcher

09/2015 - 02/2022

- Proposed an estimation method for nonlinear regression with theoretical backgrounds and accomplished 65% (with time series) and 41% (with spatial data) improvements in RMSE, resulting in 3 publications (link1, link2, link3).
- Applied semi-parametric quantile regression to model ion concentration distributions in groundwater, creating a nationwide water contamination map for South Korea, resulting in 2 publications (paperlink & manuscript).

EDUCATION

Seoul National University

Seoul, South Korea

Ph.D., Statistics (3.96)

09/2015 - 02/2022

Seoul National University

Seoul, South Korea

B.S., Industrial Engineering & Statistics (3.83)

03/2010 - 08/2015

SKILLS

- **Knowledge:** Machine Learning, Deep Learning, Data Science, Computer Vision, Natural Language Processing (NLP), LLM, VLM, RAG, Interpretable ML & AI, Multi-task learning, Prompt Engineering, Reinforcement Learning.
- **Programming Skills:** Python (PyTorch, Tensorflow, Transformers, Spark, Deepspeed, Llama-factory), C++, SQL, R.
- **Technical Skills:** Docker, Airflow, Git, VectorDB, Shell Scripting, High Performance Computing, Slurm.

EXTRACURRICULAR ACTIVITY

Machine Learning Seminar

Seoul, South Korea

Seminar Lead

01/2019 - 11/2023

- Organized weekly ML seminar covering 4 books, 2 courses, 2 hackerthons and 100+ articles on DL, generative AI, reinforcement learning, meta learning and others.